CASE

Margaret, 68 · sudden right-arm weakness and slurred speech 90 minutes ago · NIHSS 12 · CT: no hemorrhage · CT angiogram: **left middle cerebral artery occlusion** · **acute ischemic stroke**

BIO 004 · WEEK 7 · CASE DEEP DIVE

Functional Organization & Nervous Tissue

A 68-year-old who suddenly can't lift his right arm. A clot in one cerebral artery. Neurons dying by the millions per minute.

PATIENT CASE

Margaret is 68, in atrial fibrillation (off her anticoagulant). Ninety minutes ago she developed **sudden weakness of her right arm and face** and slurred speech.

In the ED her NIHSS is 12 (significant deficit). CT shows no hemorrhage. CT angiogram reveals a **clot in her left middle cerebral artery**. The stroke team activates the thrombectomy protocol. The diagnosis is **acute ischemic stroke from a cardioembolic source**.

Every minute of cerebral ischemia kills ~2 million neurons. Margaret's case rests on knowing what a neuron is, what glia do, and how the CNS is organized so that one clot in one artery produces this specific pattern of loss.

> **GUIDING QUESTION**
>
> How is the nervous system divided, what are the two cell families in nervous tissue, and how does the architecture of gray matter and white matter produce focal deficits when one artery occludes?

PREWORK

Companion to your Week 7 Functional Organization & Nervous Tissue prework page.

1 CNS VS PNS

Two anatomic divisions, multiple functional ones

Central nervous system (CNS)
Brain + spinal cord. Wrapped in meninges and bathed in CSF.

Peripheral nervous system (PNS)
Everything outside the CNS: cranial nerves, spinal nerves, peripheral nerves, ganglia, autonomic plexuses.

Functional divisions: **somatic** (voluntary motor + sensory from skin/joints/muscle) and **autonomic** (involuntary control of viscera, split into sympathetic and parasympathetic). Sensory side = *afferent*; motor side = *efferent*.

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2 THE NEURON

Signaling cell with three regions

Dendrites
Receive signals from other neurons. Often heavily branched.

Cell body (soma)
Contains nucleus, organelles. Site of protein synthesis. Integrates inputs.

Axon
Single long process. Conducts the action potential away from the soma to other neurons, muscles, or glands. Can be very long (motor neuron axons run from spinal cord to foot).

Functional classification: **sensory (afferent)** neurons carry signals to CNS; **motor (efferent)** neurons carry signals from CNS to effectors; **interneurons** connect the two within the CNS.

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3 NEUROGLIA

The support cells that outnumber neurons

Six types of glia, four in the CNS and two in the PNS:

- **Astrocytes (CNS)**
  Most abundant. Maintain blood-brain barrier, regulate extracellular environment, support neurons, form glial scar after injury.
- **Oligodendrocytes (CNS)**
  Produce myelin in the CNS. One cell wraps multiple axons.
- **Microglia (CNS)**
  Immune cells of the brain. Phagocytose pathogens and debris.
- **Ependymal cells (CNS)**
  Line ventricles and central canal. Produce CSF (in choroid plexus).
- **Schwann cells (PNS)**
  Produce myelin in the PNS. One cell wraps one segment of one axon.
- **Satellite cells (PNS)**
  Surround neuron cell bodies in ganglia. Support and regulate environment.

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4 ACTION POTENTIAL

All-or-nothing electrical signal

Resting neurons hold a negative membrane potential (~-70 mV) by pumping ions and being selectively permeable. When stimulation depolarizes the membrane to threshold (~-55 mV), voltage-gated Na+ channels open → rapid depolarization to +30 mV → K+ channels open, Na+ channels inactivate → repolarization → brief hyperpolarization → return to rest. All-or-nothing: every action potential is the same size; intensity is encoded by frequency.

Myelin dramatically speeds conduction. Myelinated axons conduct by *saltatory conduction*: action potential jumps from one Node of Ranvier to the next, skipping the insulated internodes. Unmyelinated axons conduct slowly along the whole membrane.

5 SYNAPSE

Where neurons talk

At a chemical synapse, action potentials arriving at the axon terminal open voltage-gated Ca++ channels. Ca++ triggers vesicles containing **neurotransmitter** to fuse with the membrane and release into the synaptic cleft. Neurotransmitter binds receptors on the postsynaptic membrane → ion channels open or close → postsynaptic potential (excitatory or inhibitory).

The postsynaptic neuron integrates many inputs (some excitatory, some inhibitory). If the sum at the axon hillock crosses threshold, it fires.

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6 GRAY VS WHITE

Where cell bodies vs axons live

Gray matter contains neuron cell bodies and dendrites (where signals are received and processed). **White matter** contains myelinated axons (where signals travel). In the brain, gray matter is on the outside (cerebral cortex) and in deep nuclei; white matter is in between. In the spinal cord, the arrangement reverses — gray matter is in the H-shaped center, white matter surrounds it.

> **Why Margaret's deficit is specific.** The *left middle cerebral artery* supplies the lateral surface of the left cerebral hemisphere: most of the frontal, parietal, and temporal lobes. This includes the primary motor cortex for face and upper extremity (the "lateral homunculus") and Broca's area (motor speech) in dominant hemispheres. Occlusion produces right-sided face and arm weakness (contralateral motor cortex) and expressive aphasia (Broca's) — exactly Margaret's presentation.

✎ **DRAW ZONE 1** -- *Sketch a neuron with all parts labeled (dendrites, soma, axon hillock, axon, myelin sheath, Nodes of Ranvier, axon terminals). Add a synapse showing pre- and postsynaptic membranes.*

Star the location of action potential generation (axon hillock).

✎ **DRAW ZONE 2** *-- Diagram the cerebral hemispheres from above. Color the territory of the left middle cerebral artery and mark Margaret's deficit on the contralateral body (right face and arm).*

Note the dominant-hemisphere language areas (Broca, Wernicke) along the MCA distribution.

RETURNING TO MARGARET

Time is brain

Every minute matters. Margaret's clot is killing the neurons that supply motor and language function to half her body.

"*Right face and arm weakness*"

The motor cortex is contralateral (left brain controls right body). Face and arm are represented on the lateral surface of the precentral gyrus — exactly the MCA distribution.

"*Slurred speech*"

Broca's area (left frontal, dominant hemisphere in 95%+ right-handed people) is in MCA territory. Expressive aphasia and dysarthria result.

"*Why CT shows no hemorrhage but no obvious infarct*"

Acute ischemic stroke is invisible on CT in the first hours. CT mainly rules out hemorrhage to determine eligibility for thrombolytics.

"*Atrial fibrillation as the source*"

Stagnant blood in the left atrial appendage clots and embolizes. Cardioembolic strokes are typically large and disabling, often in MCA distribution.

"*Why thrombectomy now*"

Mechanical clot removal from large-vessel occlusions within 24 hours dramatically reduces disability. The penumbra (ischemic but not yet dead tissue) can be salvaged if perfusion is restored in time.

"*Recovery dynamics*"

Some recovery from edema and reperfusion in days. Slower recovery over months from neuroplasticity. Lost neurons don't regenerate; surviving neurons take over some functions.

Neuron, axon, gray, white. Margaret's deficits are the nervous system map showing you which neurons are starving.